PS 5 · THE FAKE HR EMAIL

C U S T O D I

Think before you trust.

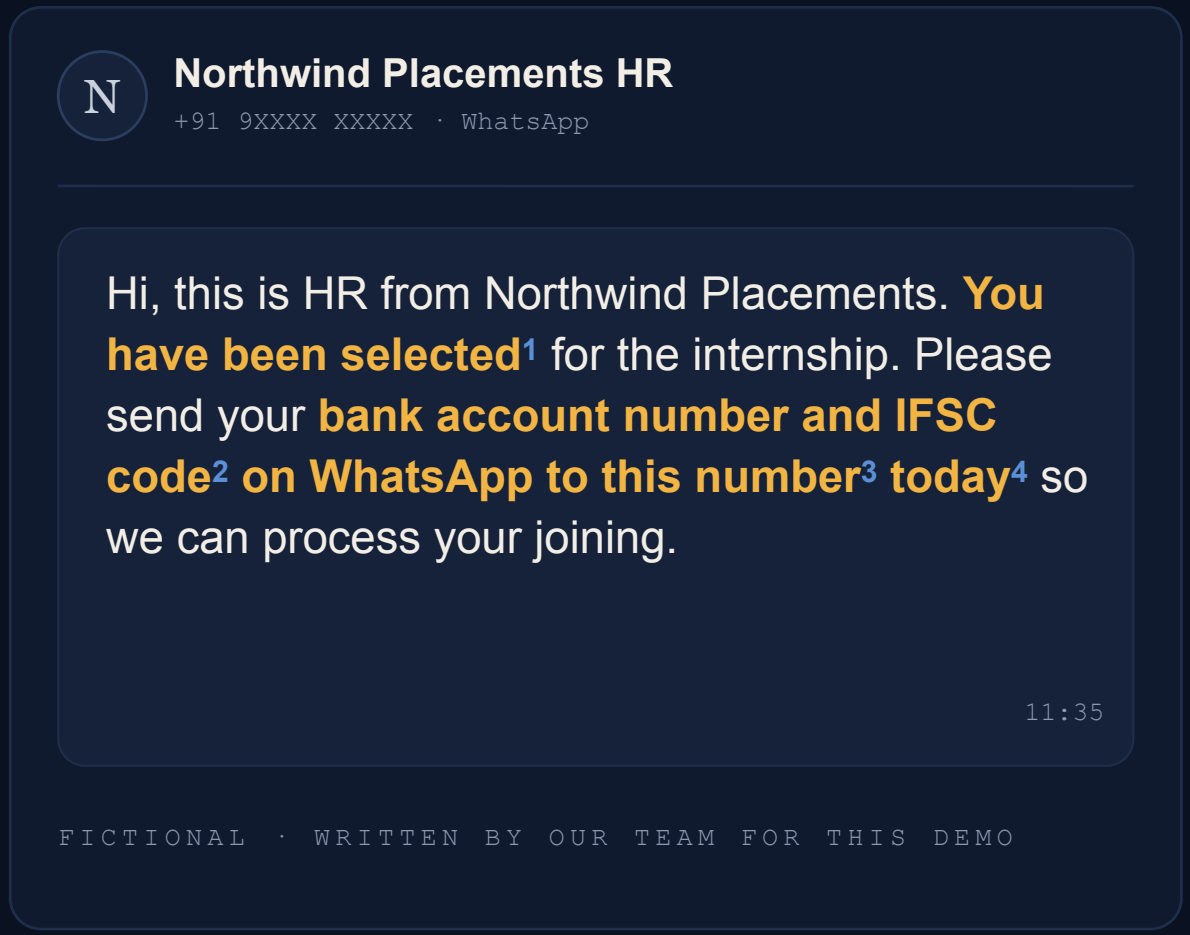
Verify before you act.

An on-device safety check for placement, internship and recruiter messages — and the links inside them.



ARTHAKRAM RAPID PROTOTYPING HACKATHON · TEAM CUSTODI

It doesn't look like a scam. It looks like good news.



HOW IT WORKS ON A STUDENT

1 **THE LURE**

"You have been selected"

A reward with no test, interview or process.

2 **THE ASK**

Bank account + IFSC

Financial details, before any offer is accepted.

3 **THE CHANNEL**

"On WhatsApp to this number"

No official record, and nobody to check with.

4 **THE CLOCK**

"Today"

No time to ask the placement cell first.

Same message. Two very different next steps.

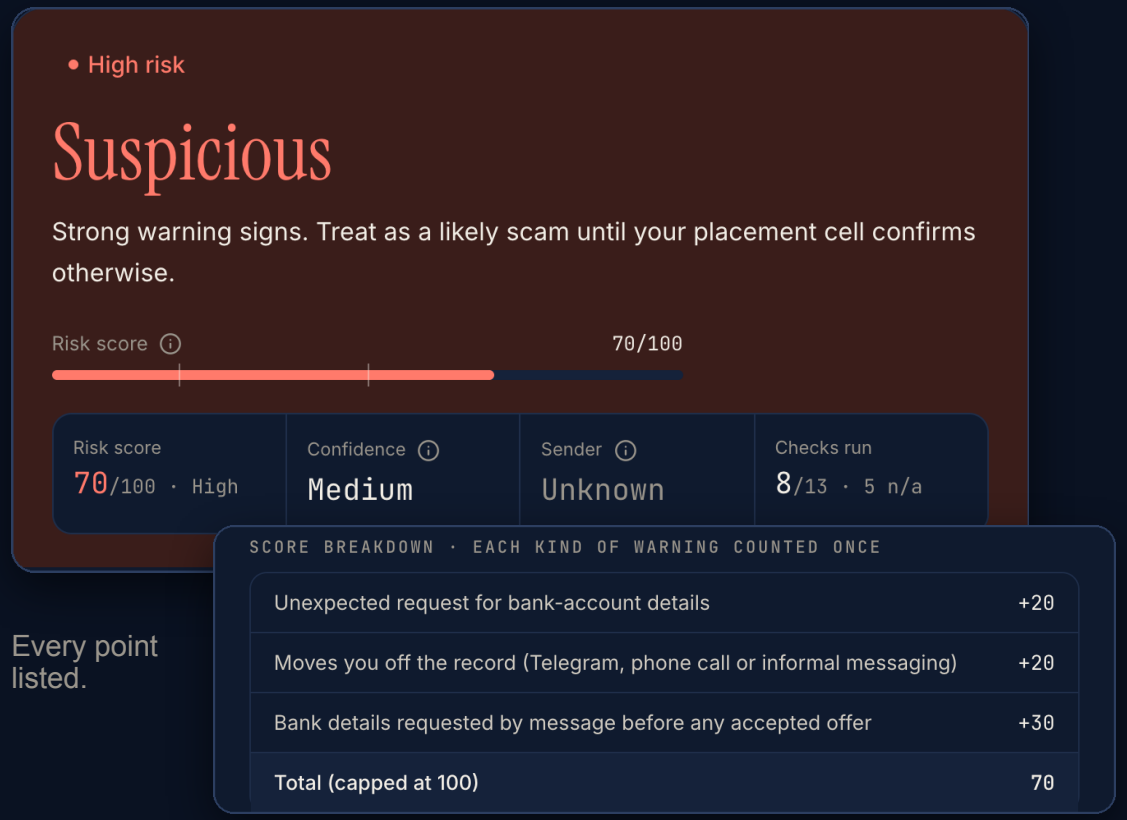
WITHOUT CUSTODI

- *“It says I'm selected.”*
- *“They need my bank details today.”*
- *“I'll just send them on WhatsApp.”*

WHAT HAPPENS

Account details sent over an unrecorded chat — to someone nobody verified.

WITH CUSTODI · REAL RESULT, SAME MESSAGE



Every point listed.

One analyzer. Three ways to reach it.

01 Scan a message

Email, SMS, WhatsApp or any text. Thirteen checks, with the exact words quoted.

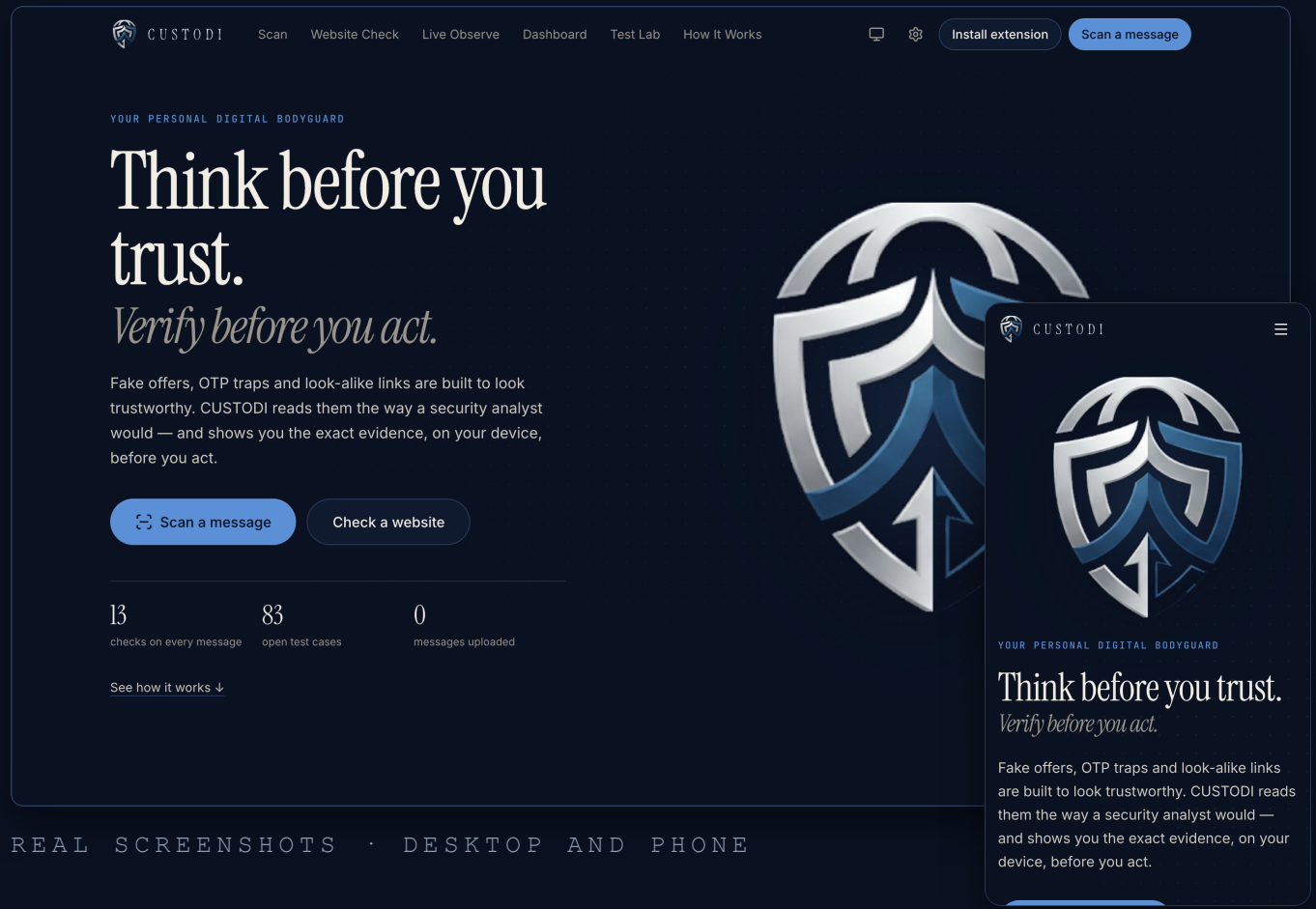
02 Check a website

Reads the address like an analyst — without opening it.

03 Check the page you're on

Chrome extension: one click, or a timed Live Observe session you control.

ALSO: INSTALLABLE APP (PWA) · SHARE A MESSAGE INTO IT · SCAN DASHBOARD · TEST LAB

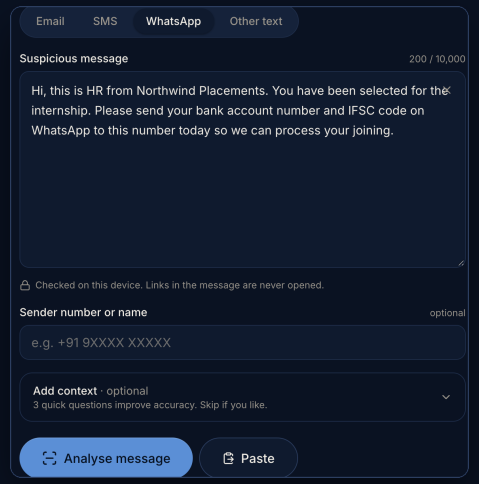


REAL SCREENSHOTS · DESKTOP AND PHONE

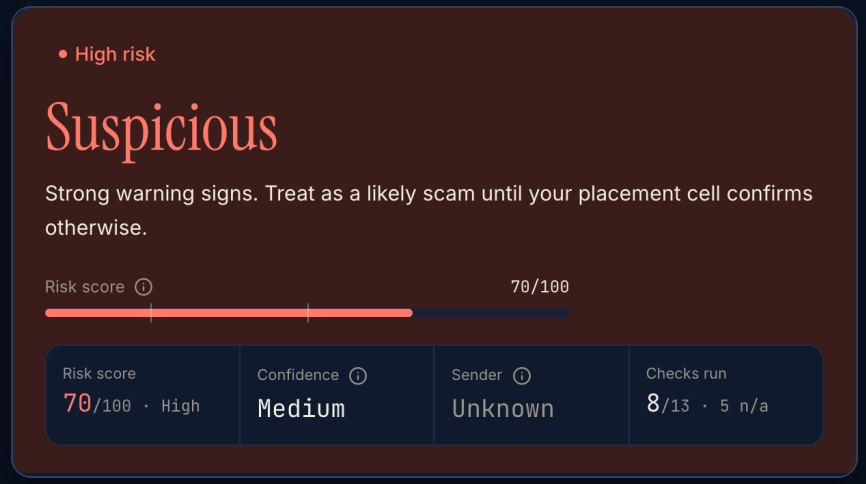
From doubt to a safe next step

1. **Gets a message** — “Selected! Send bank details today.”
2. **Opens CUSTODI** — Website, installed app, or shared in
3. **Pastes it** — Optional: sender and 3 quick questions
4. **13 checks run** — On the device. Links never opened.
5. **Score & verdict** — 70 / 100 · Suspicious
6. **Evidence shown** — Exact words quoted, in plain language
7. **Safe next step** — Verify through the placement cell

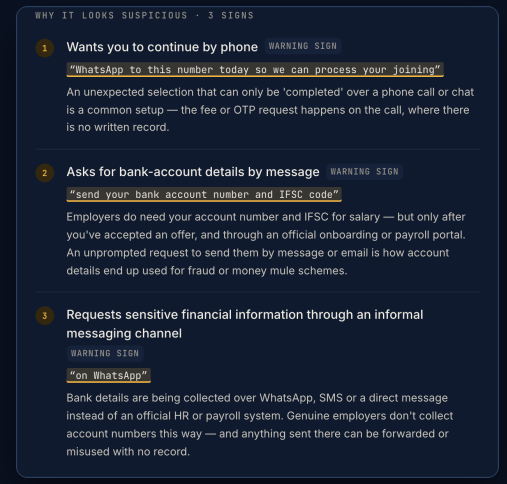
STEP 3 · PASTE



STEP 5 · VERDICT



STEP 6 · EVIDENCE



Checking a link without opening it

https://**campus-offer-verify-portal.xyz**/login

CONNECTION | REAL OWNER — THE PART THAT MATTERS | PATH

9 CHECKS · REAL RUN ON A FICTIONAL ADDRESS

●	01	Address parsing		PASSED
●	02	Hidden destination tricks		PASSED
●	03	Host type		PASSED
●	04	Look-alike & impersonation		PASSED
●	05	**Domain patterns**	“verify” + login in a .xyz address	WARNING
●	06	Connection (HTTPS)		PASSED
○	07	Your trusted list	empty — so never “Verified”	UNAVAILABLE
○	08	Threat intelligence & reputation	not connected — and it says so	UNAVAILABLE
●	09	Risk calculation		PASSED



Result: 45 / 100 · Suspicious
Next step: don't log in or pay through a link you were sent — type the official address yourself.

In Chrome, CUSTODI looks only when you ask



1 Clicks the CUSTODI icon → This page

2 Gets this tab only, at this moment (activeTab)

3 Reads page text, address and field types — not your typing

4 The same engine as the website scores it

5 Verdict, evidence; badge ! or ? on the icon

6 Decides: leave, verify, or carry on

LIVE OBSERVE · A SESSION YOU START

PROPOSED, NOT BUILT: right-click → scan selected text · Chrome Web Store

- Pick 15, 30 or 60 minutes
- Allow site access for the session
- Each page you switch to is checked
- Badge shows ON the whole time
- Timer ends or Stop now → summary; access handed back

How the engine reaches a verdict

01

Read

Message text — email, SMS, WhatsApp or other

Optional: sender and three context questions

UNTRUSTED TEXT · LINKS NEVER OPENED

02

Detect

OTP, password & PIN requests
Payment & registration fees
Urgency & threats
Secrecy & manipulation
Apps, remote access & documents
Sender indicators
Your answers
Links & look-alike domains
Reputation → “unavailable”

13 CHECKS · ENGLISH + HINGLISH RULES

03

Apply context

A warning becomes a note when it is…

1 quoted inside a scam-awareness notice

2 an OTP sent to you, with “never share”

3 bank details for an official portal — not asked back by message

NOTES SCORE 0 · STILL SHOWN

04

Score

Each warning category counts once

Hand-set weights, +5 to +35

Capped at 100

Critical override → High: OTP / PIN / password request, remote-access tool, .apk or .exe install

SCORING.TS

05

Explain

0–19 Low risk

20–49 Caution

50–100 Suspicious

Plus, reported separately:

Confidence — how much CUSTODI had to work with

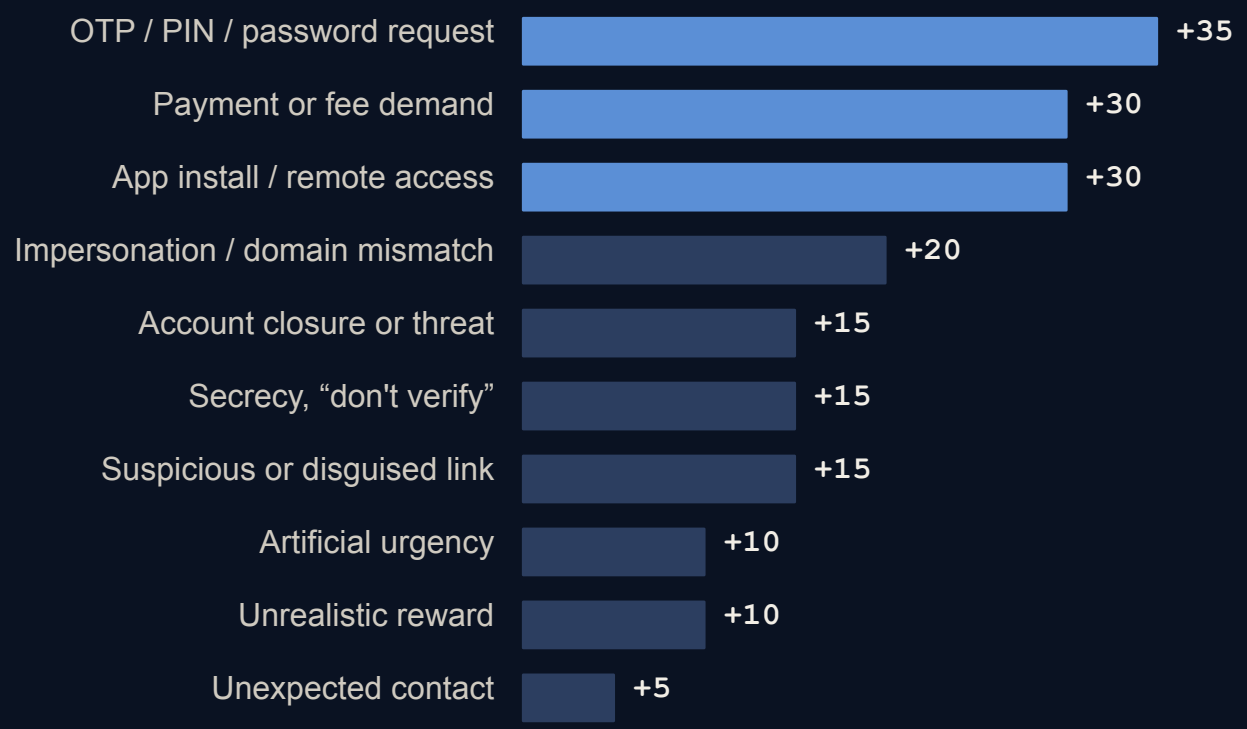
Sender — “verified” only if on your own list

Then: quoted evidence and one safe next step

VERDICT · EVIDENCE · NEXT STEP

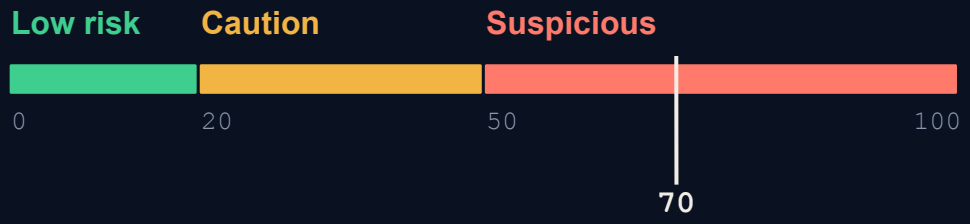
Every point is earned — and shown

WEIGHT PER CATEGORY · COUNTED ONCE



Also +20 each: ID documents over chat · a job with no assessment · bank details asked · hidden AI instructions · moving you off the record · an OTP request you report. Context: +20 for a fee in a job context, +30 for bank details by message before any accepted offer.

THE SCALE



THE WHATSAPP MESSAGE

20 + 20 + 30 = 70

+20 bank-account details requested
+20 moved off the record — “on WhatsApp to this number”
+30 asked by message, before any accepted offer

Critical override

A direct OTP / PIN / password request, a remote-access tool or an .apk / .exe install is High — whatever the total.

HAND-SET WEIGHTS, TUNED ON OUR FICTIONAL CASES — NOT A PROBABILITY OF FRAUD

Same request. Different context. Different answer.

WHATSAPP · BEFORE ANY OFFER

*"...Please send your **bank account number and IFSC code on WhatsApp to this number** today..."*

70 /100

Suspicious

+20 bank-account details requested
+20 moved to an informal channel
+30 by message, before an accepted offer

OUR DEMO MESSAGE

EMAIL · AFTER ACCEPTING · GMAIL SENDER

*"Now that you have accepted our offer, please **email us your bank account number** and IFSC for payroll setup."*

40 /100

Caution

+20 bank-account details requested
+20 personal email used for official hiring
No +30 — the offer was already accepted

TEST LAB CASE R10

EMAIL · AFTER ACCEPTING · OFFICIAL PORTAL

*"...please add your bank account number and IFSC **on the official onboarding portal**. Do not email these details."*

0 /100

Low risk

Bank request becomes a note:
directed to an official portal,
not asked back by message → 0 points

TEST LAB CASE M42

CUSTODI doesn't flag the words "bank account". It weighs who is asking, how, and when.

Built so messages never leave the device

YOUR DEVICE

CUSTODI web app

REACT · VITE · INSTALLABLE PWA

Scan · Website check · Dashboard · Test Lab · Live Observe page

Results kept in browser storage — full text off by default

App shell cached by a service worker

Chrome extension

MANIFEST V3 · POPUP + SERVICE WORKER

This page · Message · Live Observe

Results kept in chrome.storage.local

Site access only during Live Observe; talks to the website only on localhost

`@custodi/core` — one TypeScript analyzer, imported by both

13 MESSAGE CHECKS · WEBSITE & PAGE CHECKS · SCORING · 83 TEST CASES · 120 UNIT TESTS

NOTHING LEAVES

- ✖ No server
- ✖ No AI service
- ✖ No outside lookups
- ✖ No uploads
- ✖ No accounts

What CUSTODI won't pretend to know

0 messages uploaded — there is nowhere to upload them to.

- Analysed in memory, on the device
- Links read as text — never opened
- Results only, with export and delete
- Live Observe is visible, timed, and hands access back

LIMITS WE STATE OPENLY

01 Can't read inboxes — you paste or share the message in.

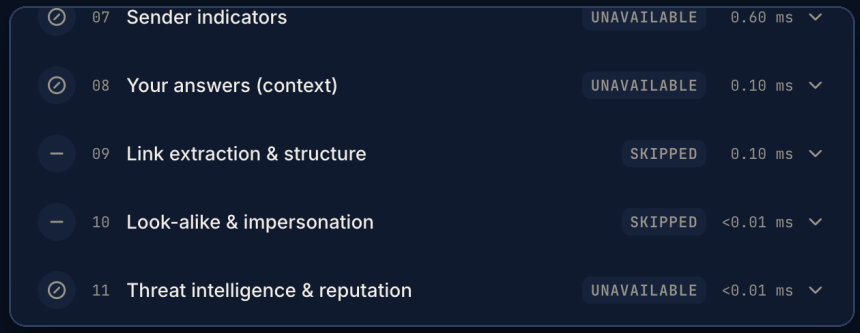
02 No blocklist, domain-age or sender-authentication lookups.

03 Rules-based: new scam wording can slip through.

04 A Low result means no signs found — never “safe”.

05 English and common Hinglish phrasing only.

06 Weights are tuned on our own fictional cases.



Missing checks are shown as unavailable — never faked.

Tested on 83 cases — 34 of them must stay calm

83/83

fictional cases behave as designed

49
RISKY · CAUGHT

34
HARMLESS · LEFT CALM

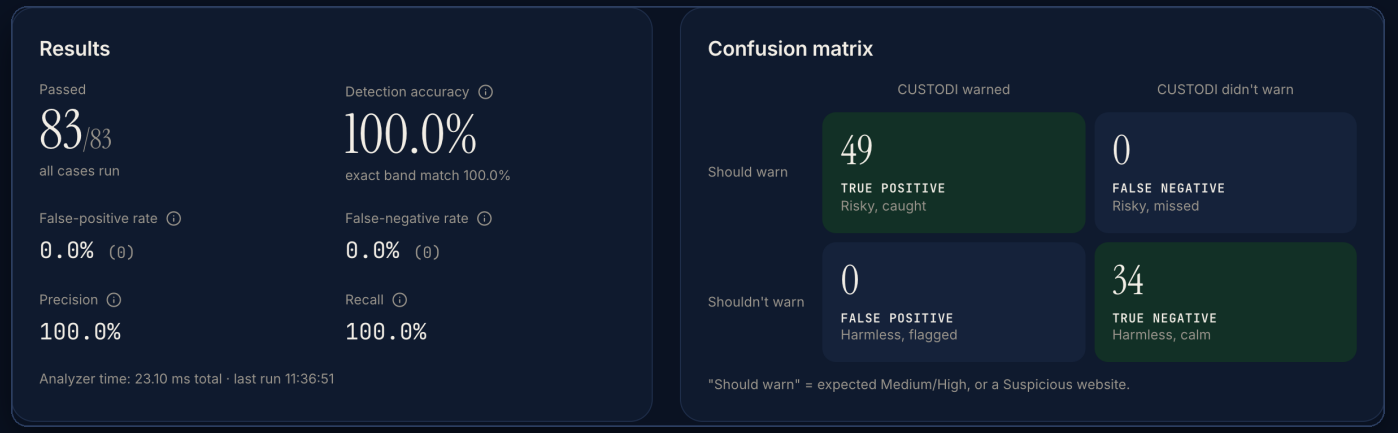
0
FALSE ALARMS

0
MISSED

120 automated unit tests pass.

A false alarm found while testing in real Chrome is now regression case R01.

OUR OWN CASES: PROOF THE ENGINE WORKS AS DESIGNED — NOT REAL-WORLD ACCURACY





What's different today, and what comes next

DISTINCTIVE TODAY · BUILT

It shows its working

Every point, every quoted word — even checks that couldn't run.

It understands context

Portal onboarding, OTP deliveries and warnings stay calm.

It's honest about gaps

Missing data is “unavailable”, and nothing is ever called “safe”.

It's private by design

One on-device engine for the website, the app and the extension.

WHAT COMES NEXT · PROPOSED, NOT BUILT

- **Now: working prototype**
  Website, installable app, Chrome extension, Test Lab.
- **Pilot with a placement cell**
  Measure on real, consented and anonymised messages.
- **Verified recruiter domains**
  The placement cell supplies the trusted list.
- **Chrome Web Store release**
  One-click install — no developer mode.
- **Opt-in reputation lookups**
  Domain age and blocklists, only with consent.
- **More Indian languages**
  Beyond English and Hinglish.



Think before you trust.

Verify before you act.

C U S T O D I

github.com/govindanidarsh/Custodi

Every example is fictional · everything runs on your device